

SECTION 1: PATIENT ENCOUNTER & CLINICAL DEMOGRAPHICS

Patient Identifier (PID) : PAT-2026-8842 Medical Record Number (MRN) : REC-8842
Patient Full Name : Eleanor Vance Patient Age / Biological Sex: 54Y / Female
Ordering Physician : Dr. Marcus Sterling, MD Accession Number : ACC-2026-90412
Collection Timestamp : 2026-08-23 08:30 AM EST Diagnostic Center : Boston Clinical Lab
Diagnostic Focus : Type 2 Diabetes Mellitus with Compounded Cardiometabolic Strain

SECTION 2: GLYCEMIC & PANCREATIC ENDOCRINE AXIS (DIABETES SCREEN)

Biomarker Name	Observed Value	Reference Range	Clinical Status
Fasting Blood Sugar (FBS)	168.0 mg/dL	70.0 - 99.0 mg/dL	ELEVATED (ADA)
Fasting Serum Insulin	28.0 uIU/mL	2.0 - 24.9 uIU/mL	HYPERINSULINEMIC
HOMA-IR (Insulin Resistance)	11.61 Index	< 1.90 Optimal	SIGNIFICANT RESISTANCE (>=3.0)
QUICKI (Sensitivity Index)	0.312 Index	> 0.382 Normal	RESISTANT

SECTION 3: CARDIOVASCULAR & HEMODYNAMIC PROFILE (CVD / ASCVD SCREEN)

Biomarker Name	Observed Value	Reference Range	Clinical Status
Blood Pressure (Systolic/Dia)	154/96 mmHg	90-119 / 60-79 mmHg	STAGE 2 HYPERTENSION
Mean Arterial Pressure (MAP)	115.3 mmHg	70.0 - 100.0 mmHg	ELEVATED VASCULAR LOAD
Pulse Pressure (Compliance)	58.0 mmHg	30.0 - 50.0 mmHg	COMPLIANT VESSEL WALLS
Total Serum Cholesterol	245.0 mg/dL	< 200.0 mg/dL	HIGH ATHEROGENIC BURDEN
Atherogenic Index Ratio	5.44 Ratio	< 4.50 Desirable	ELEVATED ATHEROGENICITY

SECTION 4: ONCOLOGICAL & SYSTEMIC INFLAMMATORY BIOMARKERS (CANCER SCREEN)

Biomarker Name	Observed Value	Reference Range	Clinical Status
Body Mass Index (BMI)	32.8 kg/m2	18.5 - 24.9 kg/m2	OBESE CLASS I+
Systemic Inflammatory (SMIL)	83.6 / 100	< 50.0 Baseline	HIGH INFLAMMATORY BURDEN
Resting Heart Rate	86 bpm	60.0 - 100.0 bpm	PHYSIOLOGICAL RHYTHM
Mitogenic Signaling Indicator	ELEVATED	Low Risk Baseline	PRO-INFLAMMATORY MITOGENIC FLUX

SECTION 5: DUAL-AI (GEMINI + OPENAI) CLINICAL CONSENSUS & ATTENDING DIRECTIVES

- Primary Diagnostic Staging: Type 2 Diabetes Mellitus with Compounded Cardiometabolic Strain
- Multi-Disease Triad Evaluation: Calibrated risk vectors evaluated via Random Forest / XGBoost & SHAP.
- Confirmatory Diagnostic Orders: HbA1c, Fractionated Lipid Panel (LDL-C/ApoB), hs-CRP, and 12-Lead ECG.
- Cryptographic Tamper Seal: Deterministic SHA-256 Hash pinned to IPFS and anchored to Sepolia Testnet.

Digitally Certified by: Dr. Marcus Sterling, MD, FCAP (License #MD-782910)
Cryptographic Audit Hash: 0x8a92f7c19b884210e5f29d74a01c38e92f1503c8 (TrustMedAudit.sol Verified)